Ngo Chi Nghia

AI Engineer | Generative AI, LLM & RAG Applications

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Professional Summary

Entry-level AI Engineer with 5 months of internship experience developing LLM-powered applications and Retrieval-Augmented Generation (RAG) systems. Proficient in Python, LangChain, LangGraph, FastAPI, and LLM API integration, with hands-on experience in hybrid retrieval, semantic search, and retrieval evaluation. Familiar with AI Agent workflows and containerized deployment using Docker and Linux. Co-author of an RAG research paper currently under review, with a second manuscript in preparation. Seeking an AI Engineer, GenAI Engineer, or LLM Engineer position.

Education

Nam Can Tho University — Can Tho, Vietnam
Bachelor of Engineering in Information Technology — GPA: 3.44/4.0

Technical Skills

Programming: Python, C/C++
GenAI & LLM: LLM applications, RAG pipelines, AI Agent workflows, prompt engineering, OpenAI API, Gemini, Groq
Frameworks & Backend: LangChain, LangGraph, FastAPI, RESTful APIs
Retrieval & Vector Search: ChromaDB, multilingual-E5 embeddings, hybrid search (dense + BM25), Reciprocal Rank Fusion (RRF)
Machine Learning & Computer Vision: Scikit-learn, TensorFlow, Hugging Face, YOLOv8, OpenCV
Deployment & Tools: Docker, Docker Compose, Nginx, Ubuntu Linux, Git, Claude Cli, OpenAI Cli

Experience

ADHIGHTECH (Tien Phong Engineering Technology Co., Ltd.) — Can Tho, Vietnam
AI Engineer/Gen AI Intern — *Mar. 2026 – Jul. 2026*

- Strengthened Python programming and data processing skills using NumPy and Pandas to support AI application development and data preparation.
- Developed and experimented with Retrieval-Augmented Generation (RAG) pipelines, exploring semantic search, BM25, hybrid retrieval, clustering-based retrieval, and LLM-based relevance filtering.
- Evaluated retrieval and answer quality using information retrieval and text generation metrics to analyze system performance and identify improvement opportunities.
- Integrated Large Language Models (LLMs) through APIs, managed API key configurations, and applied prompt engineering to improve response relevance and output quality.
- Developed RESTful APIs using FastAPI and configured containerized environments with Docker and Docker Compose on Ubuntu Linux.
- Used Claude Code CLI and OpenAI Codex CLI to assist with code development, debugging, error analysis, and refactoring.
- Explored AI Agent workflows in Generative AI applications, including tool usage, task execution, and integration with LLM-powered systems.

Selected Projects

ERC-RAG – Energy-Based Retrieval Clustering for RAG | GitHub | Portfolio — 2026

- Addressed limitations of cosine similarity-based retrieval for Vietnamese economic question answering by proposing an energy-based, cluster-level retrieval framework that evaluates groups of candidate documents rather than individual documents alone.
- Implemented a two-stage retrieval pipeline using multilingual-E5 embeddings and cosine similarity to retrieve the top 40 candidate documents, followed by K-Means clustering with silhouette-based cluster selection and Energy Distance-based cluster ranking.
- Applied LLM-based relevance filtering to select query-relevant documents from the winning cluster before providing contextual evidence for answer generation.
- Evaluated the framework on the Vietnamese Economy Information Database (VEID) benchmark using 120,165 document chunks and 1,000 question-answer pairs, achieving NDCG@5 of 0.83, MRR of 0.62, and Hit@5 of 0.64 for retrieval, alongside ROUGE-2 F1 of 0.68 and BLEU-2 of 0.65 for answer generation.

Vietnamese Legal Document RAG Web Application | GitHub | Portfolio — Jan. 2026

- Developed a learning-oriented RAG application for Vietnamese legal and IT/cybersecurity documents to explore document retrieval and context-grounded question answering.
- Implemented web crawling and document processing to collect and prepare source content for the RAG pipeline.
- Integrated document retrieval and LLM-based answer generation, using LangGraph to orchestrate the workflow.
- Practiced application deployment with Docker Compose and Nginx in an Ubuntu Linux environment.

PolyAqua Agent – AI Assistant for Shrimp and Crab Farming | GitHub | Portfolio

- Developed an AI assistant prototype to explore disease recognition in shrimp and crabs and provide relevant farming guidance.
- Integrated YOLOv8n-cls for image classification with a RAG pipeline combining BM25 and semantic search for knowledge retrieval.
- Combined computer vision and LLM-based generation to provide context-aware suggestions for potential aquaculture issues.

Publications & Awards

Research Papers

- *ERC-RAG: An Energy-based Retrieval Clustering for RAG in Vietnamese Economy Chatbots* | Paper (Under Review)
- *Multi-Query ERC-RAG: An Energy-based Retrieval Clustering Framework for Retrieval-Augmented Generation* (Manuscript in Preparation)

Finalist Certificate – FPTU Can Tho Olympic AI 2026.

FPT Software Talent Assessment – Certificate of Participation (Sep. 2026).

Languages

English: VSTEP B1 | **Vietnamese:** Native